Hair Care Product Online Store System

Project Type: Mini Database Design Project

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Introduction

Overview

Creating a database system for an online care store involves creating a structured schema to manage various aspects such as products, customers, orders, administrators and shipping.

Objectives

This system contains an interface that is easy to use by customers and achieves high efficiency. And to be fast performance and products get high ratings and become one of the favorite systems for customers. And achieves high productivity.

Scope

- Open and start the system quickly.
- Contains enough details for all different hair products at affordable prices.
- Doesn't take much time to make the orders.
- Fast delivery of orders to the customer with high quality.

System Goal

- Shop from anywhere, anytime) All users can use this system anytime and from anywhere in the kingdom of Saudi Arabia).

- Avoid traffic and parking hassles.
- Access to a vast array of brands and products.
- Ability to easily compare prices across retailers.
- Easier to find specialized items for specific hair types.
- Often lower prices due to reduced overhead costs.

Functional Requirements:

- Store products details and data of users, orders, shipping and administrator.
- Display all products to users in an organized form.
- Customer can get order history by using his/her id and update or delete it if wants by using order number.

A **mini database system** refers to a lightweight or simplified database management system (DBMS) that typically handles smaller amounts of data or simpler applications compared to full-scale, enterprise-level systems. These are often used for small projects, learning purposes, or embedded systems where resources like memory and processing power are limited.

Methodology

Detailed Description:

- User Requirements Gathering: Gather the requirements of the users by discussing with potential users to understand what the users' needs and expectations from the system are.

- ERD and Use Case Diagrams: Draw an ERD and use case diagrams representing all the entities in the system along with their inter-relationships in a pictorial way to make the structure of the system very clear.

- Implementing the Database Using MySQL: Using MySQL, implement the actual code for the database, applying its accuracy and efficiency in handling data about products, customers, and orders.

- Running the Code and Testing: Run the code to ensure everything is working as it should and the database and all of its functions behave as expected.

- System Implementation: The system is implemented to enable users to navigate through and order hair care products easily, giving a seamless, user-friendly experience while shopping.

Tools & Technology:

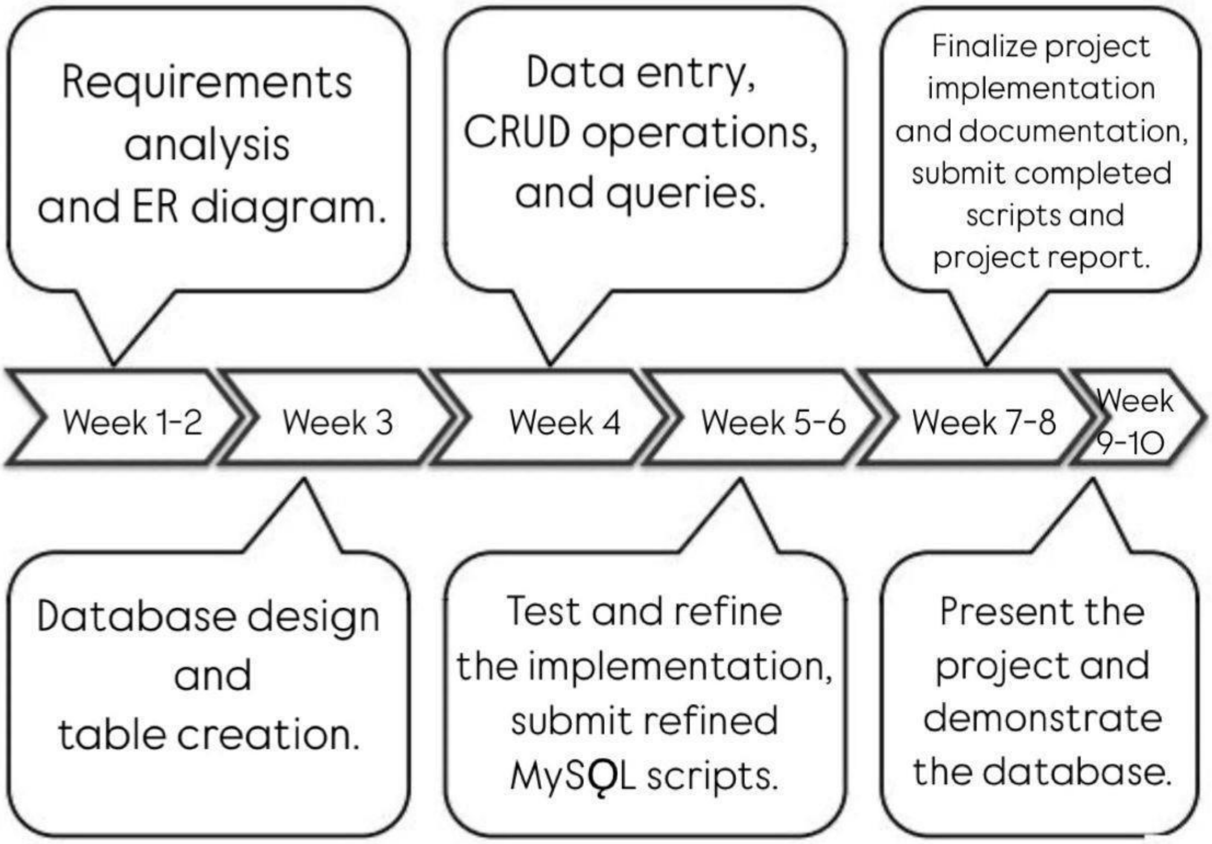
Database Management System (DBMS): MySQL

MySQL: is one of the most popular open-source relational database management systems (RDBMS). It uses structured query language (SQL) for database management, and it is known for its speed, reliability, and ease of use. It's widely used for web applications and can handle large amounts of data.
Designing Tool: visual paradigm
Visual Paradigm: is a powerful design and modeling tool used for software development, business process modeling, and system architecture design. It supports a wide range of design methodologies, including UML (Unified Modeling Language).
UML Diagrams: It supports various UML diagrams like class diagrams, sequence diagrams, activity diagrams, and use case diagrams to visually represent the structure and behavior of a system.
SQL editors: MySQL Workbench
MySQL Workbench: is a popular integrated development environment (IDE) for MySQL databases. It provides a suite of tools for database design, querying, and management.

Some of its key features include:

- **SQL Development**: A powerful SQL editor with syntax highlighting, auto completion, and error highlighting. You can write, edit, and execute SQL queries easily.
- **Database Design**: Tools for designing and visualizing databases using Entity-Relationship Diagrams (ERD).
- **Data Modeling**: You can create and modify databases, tables, and relationships visually.
- **Server Administration**: Provides tools for managing MySQL server settings, user accounts, and backups, among other things.
- **Query Management**: It offers a query history feature and allows for managing and executing multiple queries at once.
- **Performance Monitoring**: You can monitor server performance and see real-time stats on queries and resource usage.
- **Cross-Platform**: Available for Windows, macOS, and Linux.

Timeline and Milestones

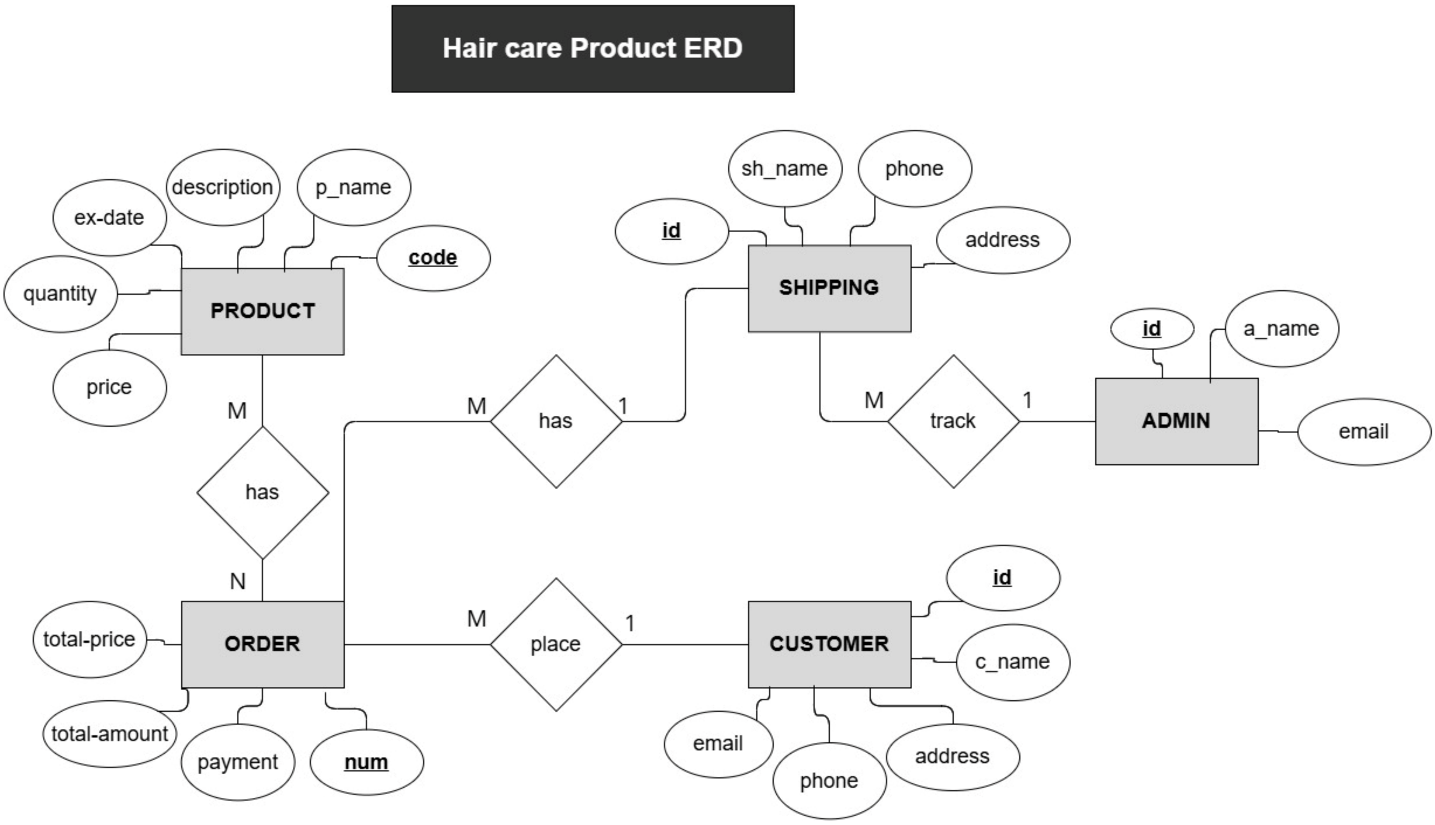


Implementation

Database Design

ERD

An ERD (Entity-Relationship Diagram) is a visual representation of the entities in a system and their relationships to each other. It helps to organize data and is often used in database design.



1. **Entities:** (product, shipping, admin, order, customer).
2. **Relationships:** These are associations between entities, typically verbs like "place," "has," or "track".
3. **Attributes:** These provide more information about entities.

Database Schema

A database schema includes tables, columns, data types, constraints, and relationships between tables.

Table 1 Product table.

Column Name/Attributes	Data Type	Description
P_code	Int	Primary key
P_name	Varchar(50)	
P_desc_	Varchar(50)	
Price	Decimal	
Quantity	Int	
Ex_date	Date	

Table 2 Customer table

Column Name/Attributes	Data Type	Description
C_id	Int	Primary key
C_name	Varchar(30)	
address	Varchar(30)	
phone	Varchar(20)	
email	Varchar(20)	

Table 3 Admin table

Column Name/Attributes	Data Type	Description
a_id	Int	Primary key
a_name	Varchar(30)	
email	Varchar(20)	

Table 4 Order table

Column Name/Attributes	Data Type	Description
O_num	Int	Primary key
Total_price	Int	
Total_amount	Int	
Payment	Varchar(20)	
C_id	Int	foreign key(customer)
Sh_id	Int	Foreign key(shipping)

Table 5 Ordered_Products

Column Name/Attributes	Data Type	Description
O_num	Int	Foreign key(order)
P_id	Int	Foreign key(product)

Table 6 shipping table

Column Name	Data Type	Description
Sh_id	Int	Primary key
Sh_name	Varchar(30)	
Phone	Varchar(20)	
Address	Varchar(30)	
A_id	Int	foreign key(Admin)

Use Case

A use case diagram is a visual representation of the functional requirements of a system, focusing on the interactions between users (actors) and the system itself. It provides an overview of the system's behavior from the perspective of its users.

1. **Actors**: These are the external entities that interact with the system. Customer, Admin, Shipping
2. **Use Cases**: These are the actions that the system provides. Customer (login, browse products, select products, make an order, select payment method)
   Admin (login, track shipment delivery)
   Shipping (receive orders, deliver orders to customer)
3. **System Boundary**: A rectangle that defines the scope of the system.
4. **Relationships**: These show how actors interact with use cases (include, extend).

Use Case Description

Table 7 Login

Use case	Login
Purpose	Allow users to login to their accounts
Require	Registration or signup
Event flow	1. Customer navigates to login page 2. Customer enters user name and password 3. Customer clicks the login button 4. Customer redirected to the home page if he is valid
Success	Customer login to the system successfully

Table 8 Browse products

Use case	Browse products
Purpose	Allow users to view available products
Require	User must be logged in
Event flow	1. Customer navigates to product page 2. System displays the list of products 3. Customer can select a specific product
Success	Customer views products successfully

Table 9 Select products

Use case	Select products
Purpose	Allow users to select product for purchase
Require	User must be logged in and browsed products
Event flow	1. Customer selects a required products 2. System displays the details of products 3. User adds products to his cart
Success	Customer have a cart with products

Table 10 Make an order

Use case	Make an order
Purpose	Allow users to place an order for selected products
Require	User have a cart with selected products
Event flow	1. Customer reviews his cart 2. Customer clicks the checkout button 3. Customer selects the payment method 4. Customer confirms his order
Success	Order is successfully placed

Table 11 Track shipment delivery

Use case	Track shipment delivery
Purpose	Allow Admin to track the shipment of orders
Require	Admin must have an account
Event flow	1. Admin views the details of orders 2. Admin contacts the shipping to track the orders 3. Admin changes the status of the orders
Success	Admin tracks all orders successfully

Table 12 Receives order

Use case	Receives order
Purpose	Allow shipping to receive all orders
Require	Admin passes incoming orders to the shipping
Event flow	1. Admins passes orders to the shipping 2. Shipping views all details 3. Shipping prepares to deliver the orders to the customer
Success	Shipping receives all orders successfully

Table 13 Deliver orders to customers

Use case	Deliver orders to customers
Purpose	Shipping delivers the orders to customers
Require	Orders must be processed and ready for shipment
Event flow	1. Shipping receives order's details 2. Shipping prepares the orders 3. Shipping contact with customers 4. Customers receives the orders
Success	Order is successfully delivered to customer

MySQL Code

Create new schema-

```
CREATE SCHEMA haircare ;
```

Create product table-

```
CREATE TABLE product (
  p_code INT primary key,
  p_name VARCHAR(50) NOT NULL,p_desc VARCHAR(50) NULL,
  price int NOT NULL, quantity INT NOT NULL, ex_date DATE NOT NULL,
);
```

Create Order table-

```
CREATE TABLE h_order (o_num INT PRIMARY KEY, total_price INT NOT NULL,
total_amount INT NOT NULL, payment VARCHAR(20) NOT NULL,
sh_id int,  c_id int,
  CONSTRAINT ship_id FOREIGN KEY (sh_id) REFERENCES  shipping (sh_id),
  CONSTRAINT cust_id FOREIGN KEY (c_id) REFERENCES  customer (c_id));
```

Create customer table-

```
CREATE TABLE customer (
  c_id INT PRIMARY KEY, c_name VARCHAR(30) NOT NULL, address VARCHAR(30)
NOT NULL,
  phone VARCHAR(20) NOT NULL,email VARCHAR(20) NULL);
```

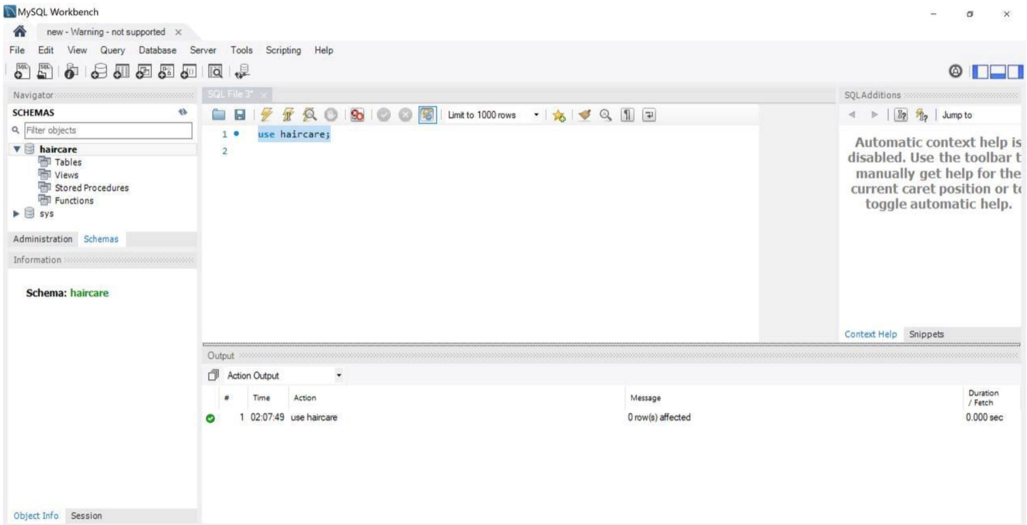
Create shipping table-

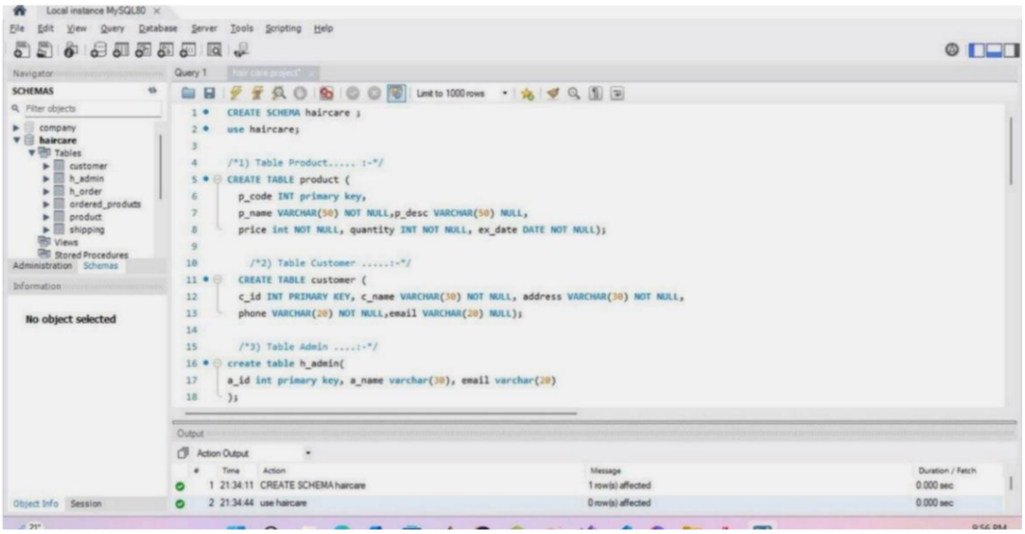
```
CREATE TABLE shipping(
  sh_id INT PRIMARY KEY, sh_name VARCHAR(30) NOT NULL, address
VARCHAR(30) NOT NULL,
  phone VARCHAR(20) NOT NULL, admin_id int,
CONSTRAINT admin_id FOREIGN KEY (admin_id) REFERENCES  admin (a_id) );
```

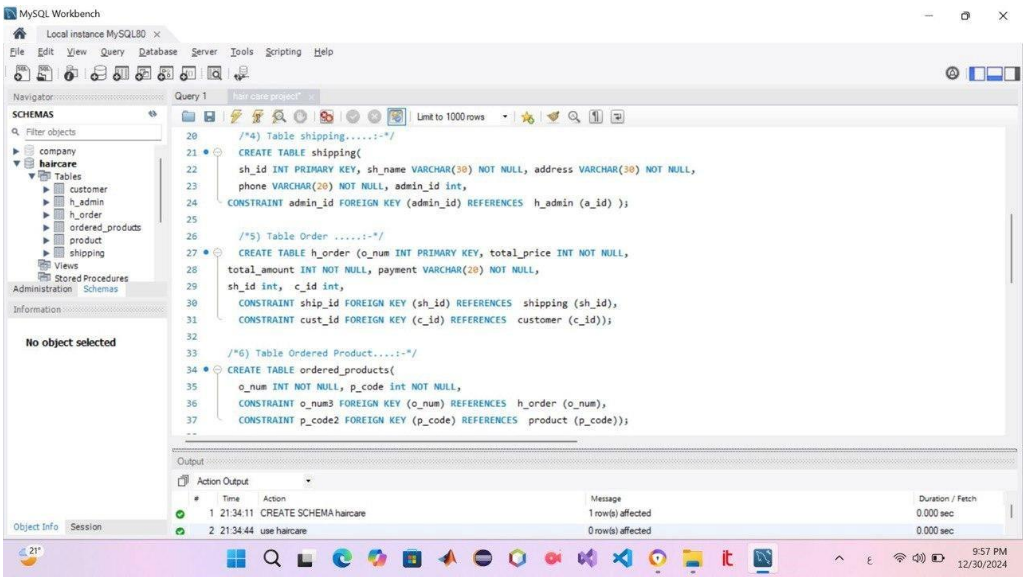
Create ordered_product table-

```
CREATE TABLE ordered_products(
  o_num INT NOT NULL, p_code int NOT NULL,
  CONSTRAINT o_num3 FOREIGN KEY (o_num) REFERENCES  h_order (o_num),
  CONSTRAINT p_code2 FOREIGN KEY (p_code) REFERENCES  product (p_code));
```

Screenshots of database and table's creation



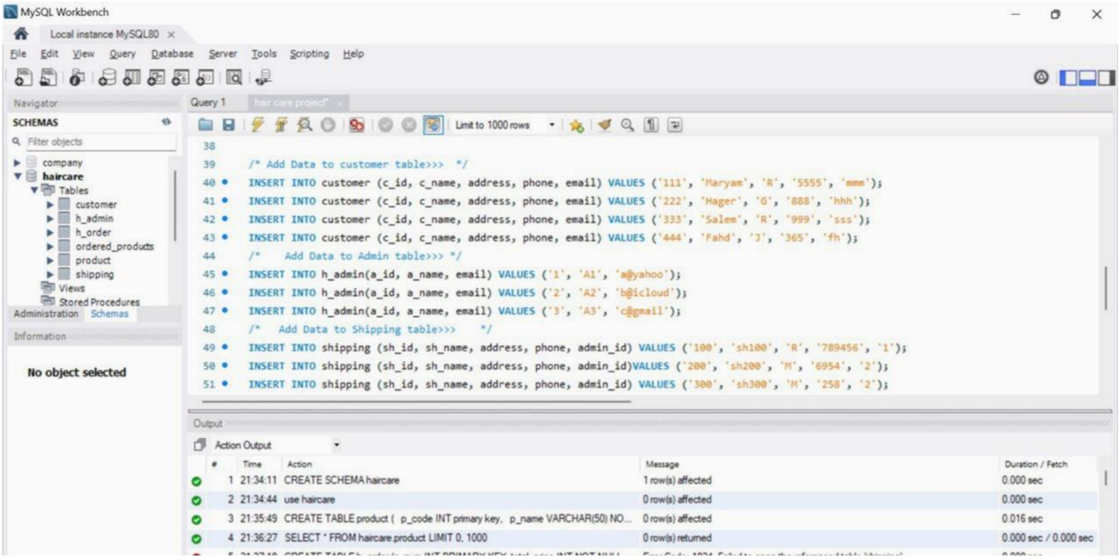


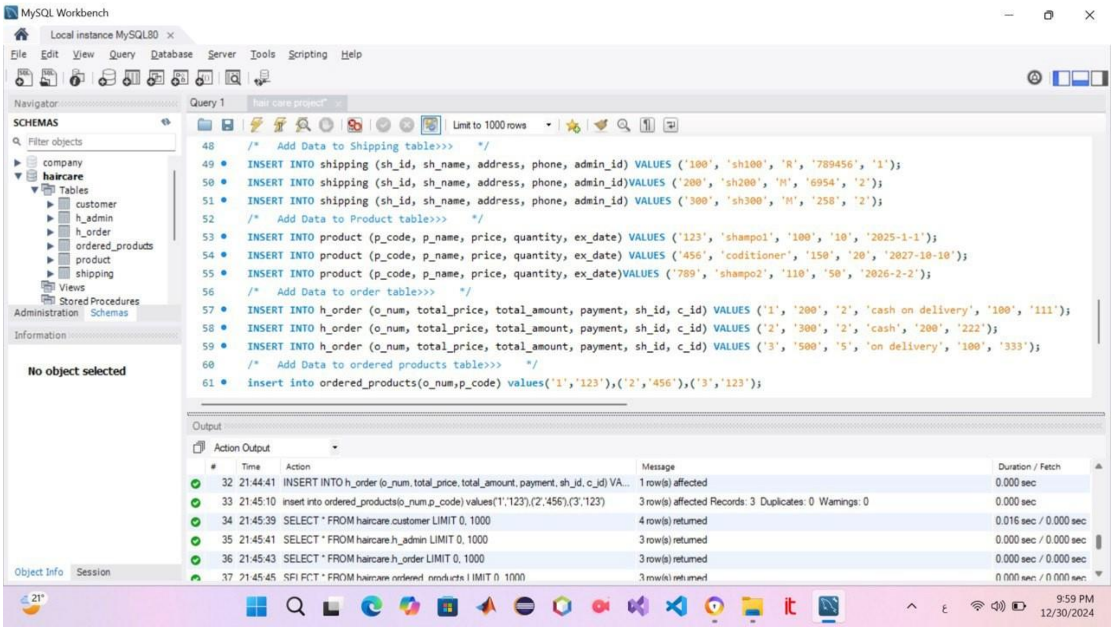


Insert data into tables

```
/* Add Data to customer table>>>  */
INSERT INTO customer (c_id, c_name, address, phone, email) VALUES ('111', 'Maryam',
'R', '5555', 'mmm');
INSERT INTO customer (c_id, c_name, address, phone, email) VALUES ('222', 'Hager',
'G', '888', 'hhh');
INSERT INTO customer (c_id, c_name, address, phone, email) VALUES ('333',
'Salem', 'R', '999', 'sss');
INSERT INTO customer (c_id, c_name, address, phone, email) VALUES ('444', 'Fahd',
'J', '365', 'fh');
/*   Add Data to Admin table>>> */
INSERT INTO h_admin(a_id, a_name, email) VALUES ('1', 'A1', 'a@yahoo');
INSERT INTO h_admin(a_id, a_name, email) VALUES ('2', 'A2', 'b@icloud');
INSERT INTO h_admin(a_id, a_name, email) VALUES ('3', 'A3', 'c@gmail');

/*  Add Data to Shipping table>>>   */
INSERT INTO shipping (sh_id, sh_name, address, phone, admin_id) VALUES ('100',
'sh100', 'R', '789456', '1');
INSERT INTO shipping (sh_id, sh_name, address, phone, admin_id)VALUES ('200',
'sh200', 'M', '6954', '2');
INSERT INTO shipping (sh_id, sh_name, address, phone, admin_id) VALUES ('300',
'sh300', 'M', '258', '2');

/*  Add Data to Product table>>>   */
INSERT INTO product (p_code, p_name, price, quantity, ex_date) VALUES ('123',
'shampo1', '100', '10', '2025-1-1');
INSERT INTO product (p_code, p_name, price, quantity, ex_date) VALUES ('456',
'coditioner', '150', '20', '2027-10-10');
INSERT INTO product (p_code, p_name, price, quantity, ex_date)VALUES ('789',
'shampo2', '110', '50', '2026-2-2');

/*  Add Data to order table>>>   */
INSERT INTO h_order (o_num, total_price, total_amount, payment, sh_id, c_id)
VALUES ('1', '200', '2', 'cash on delivery', '100', '111');
INSERT INTO h_order (o_num, total_price, total_amount, payment, sh_id, c_id)
VALUES ('2', '300', '2', 'cash', '200', '222'); INSERT INTO h_order (o_num,
total_price, total_amount, payment, sh_id, c_id)
VALUES ('3', '500', '5', 'on delivery', '100', '333');

/*  Add Data to ordered products table>>>   */ insert
into ordered_products(o_num,p_code)
values('1','123'),('2','456'),('3','123');
```

Screenshots of Insertion:





Challenges and Solutions

 Problem: Increase database volume.
Solution: Establish database connectivity with cloud computing to optimize storage and save space.

 Problem: Loss of internet connection.
Solution: Implement offline functionality that allows users to continue working until the connection is restored.

 Problem: Exchange of orders.
Solution: After confirming an order, ensure the system automatically sends a message with product details to the customer.

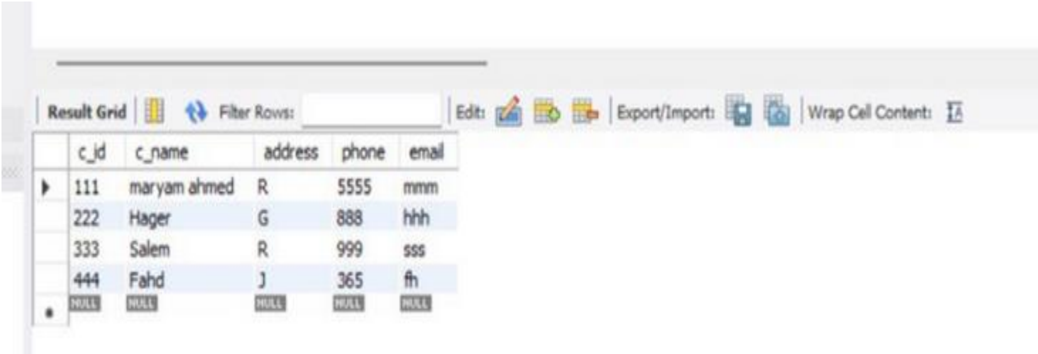
 Problem: The customer finds it difficult to use the system.
Solution: Provide an easy-to-use interface with guided windows to enhance user experience

Results

1.

/*Update table customer }}}} */

update customer set c_name='maryam ahmed' where c_id=111;



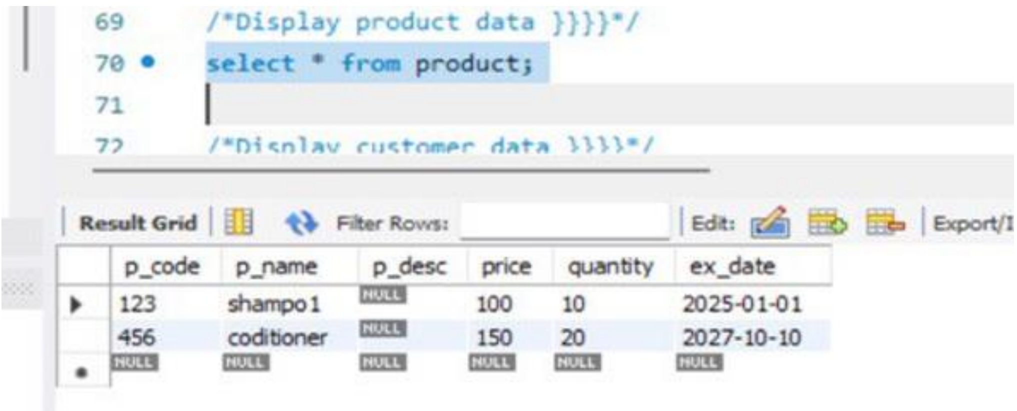
c_id	c_name	address	phone	email
111	maryam ahmed	R	5555	mmm
222	Hager	G	888	hhh
333	Salem	R	999	sss
444	Fahd	J	365	fh
NULL	NULL	NULL	NULL	NULL

The system allow the user to edit/update user information
The following example update customer name to Maryam based on the goven userid.

2.

/*Display product data }}}}*/

select * from product;



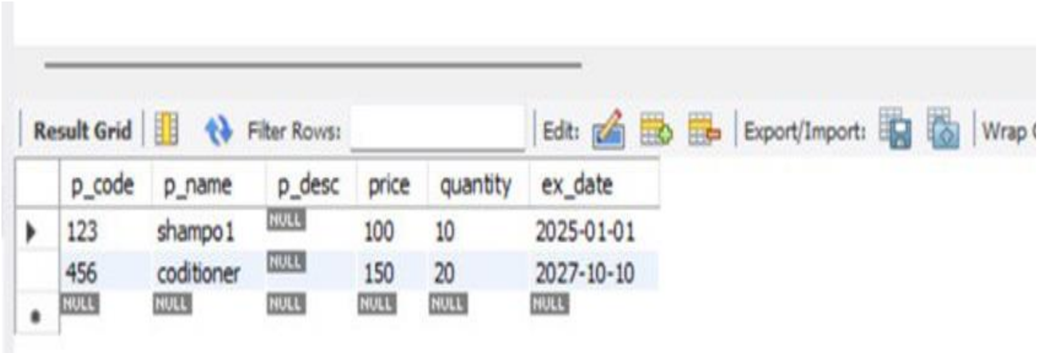
p_code	p_name	p_desc	price	quantity	ex_date
123	shampo1	NULL	100	10	2025-01-01
456	coditioner	NULL	150	20	2027-10-10
NULL	NULL	NULL	NULL	NULL	NULL

The user can browse all the available products.

3.

/*delete from table product }}}}*/

delete from product where p_code=789;



Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

p_code	p_name	p_desc	price	quantity	ex_date
123	shampo1	NULL	100	10	2025-01-01
456	coditioner	NULL	150	20	2027-10-10
NULL	NULL	NULL	NULL	NULL	NULL

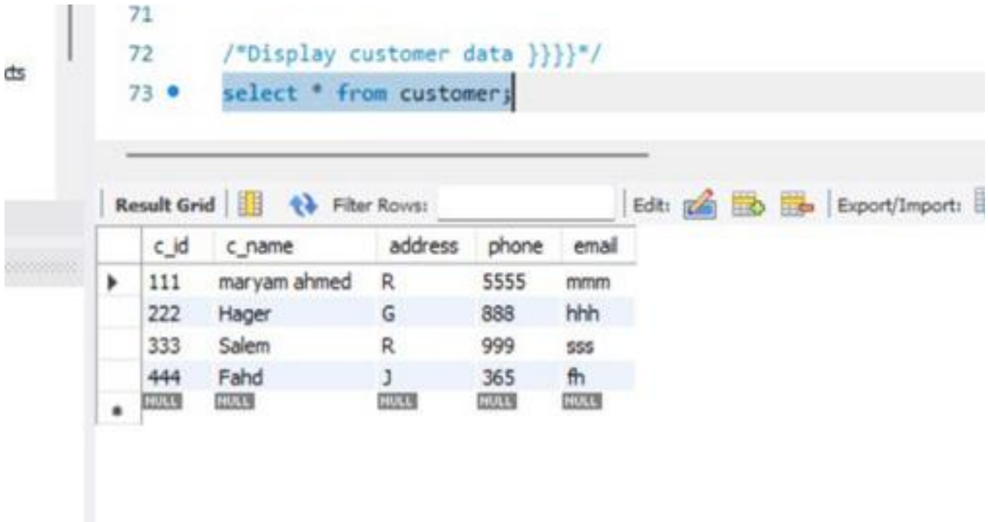
The system allow unneeded products

The following example demonstrates deleting product based on product number.

4.

/*Display customer data }}}}*/

select * from customer;



```
71
72    /*Display customer data }}}}*/
73 •  select * from customer;
```

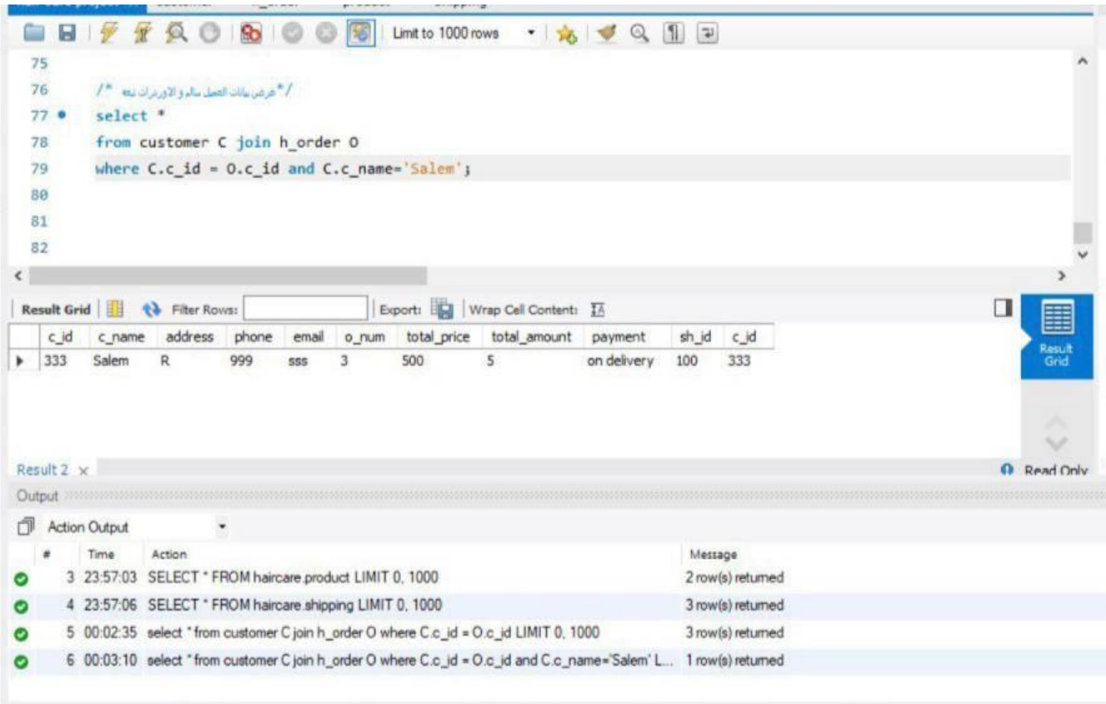
Result Grid | Filter Rows: | Edit: | Export/Import:

c_id	c_name	address	phone	email
111	maryam ahmed	R	5555	mmm
222	Hager	G	888	hhh
333	Salem	R	999	sss
444	Fahd	J	365	fh
NULL	NULL	NULL	NULL	NULL

The system provides capability to browse customers list as mentioned in the following query.

5.

select *
from customer C join h_order O
where C.c_id = O.c_id and C.c_name='Salem';



Results Overview :

The system help the customer to get information about his orders.

For example :

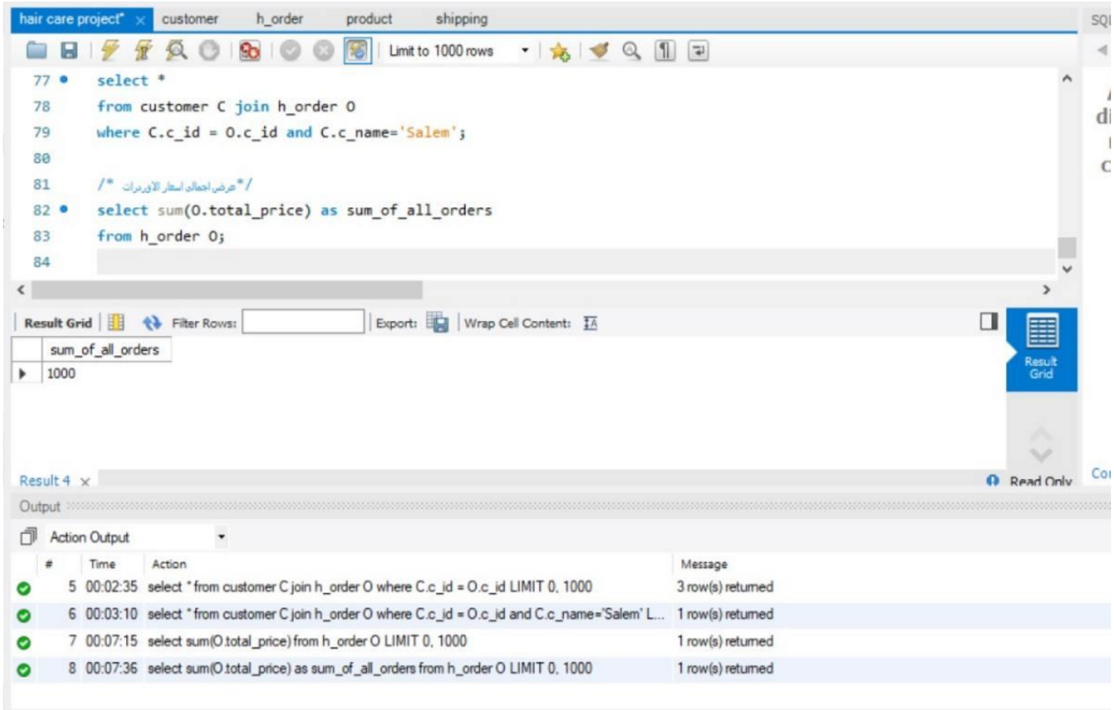
retrieves all relevant information about the customer named 'Salem' along with their associated orders .

System Benefits :

Our system provides a comprehensive view of customer activities, allowing businesses to analyze individual customer interactions. By combining customer details with their order history, stakeholders can tailor marketing strategies and enhance customer service.

6.

select sum(O.total_price) as sum_of_all_orders
from h_order O;



Results Overview :

The system provides insights into the financial performance of orders.

For example:

it calculates the total price of all orders placed within the system.

System Benefits :

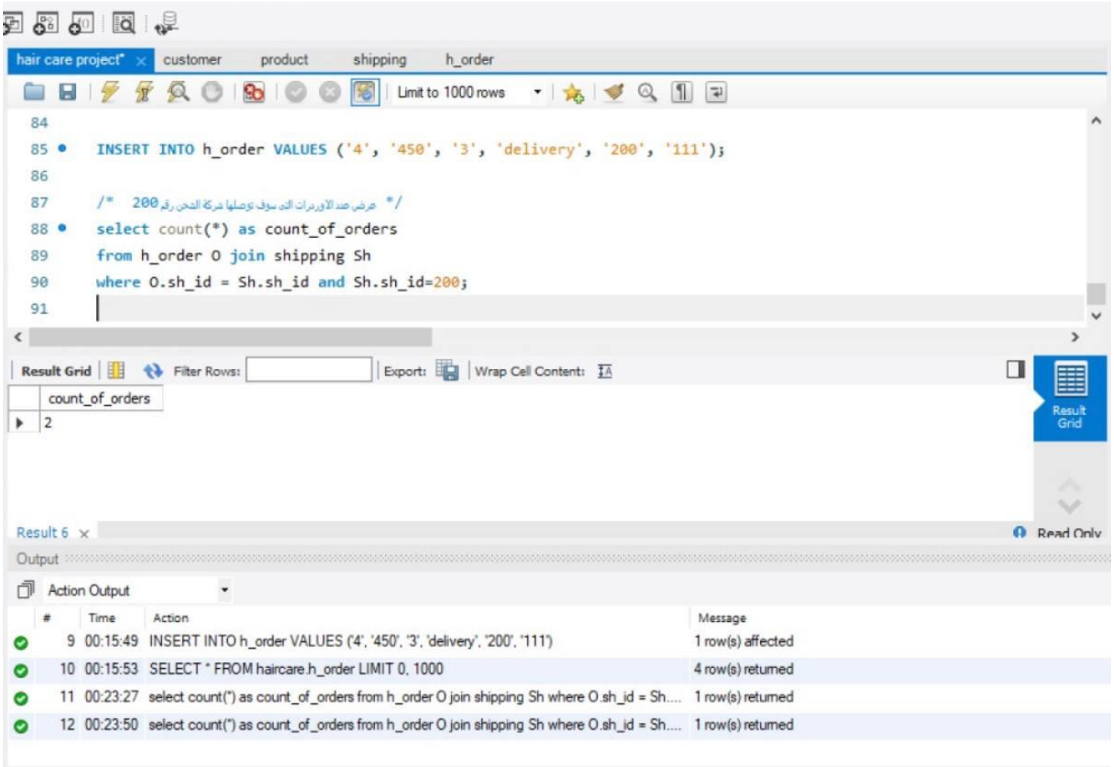
Our system enables businesses to understand their revenue streams better .

For example :

analyzing the total price of orders helps in identifying sales trends and making informed decisions about inventory and marketing strategies.

7.

select count(*) as count_of_orders
from h_order O join shipping Shwhere O.sh_id = Sh.sh_id and Sh.sh_id=200;



Results Overview :

The system tracks shipping performance and order distribution .

For example :

it counts the number of orders associated with a specific shipping ID .(200)

System Benefits :

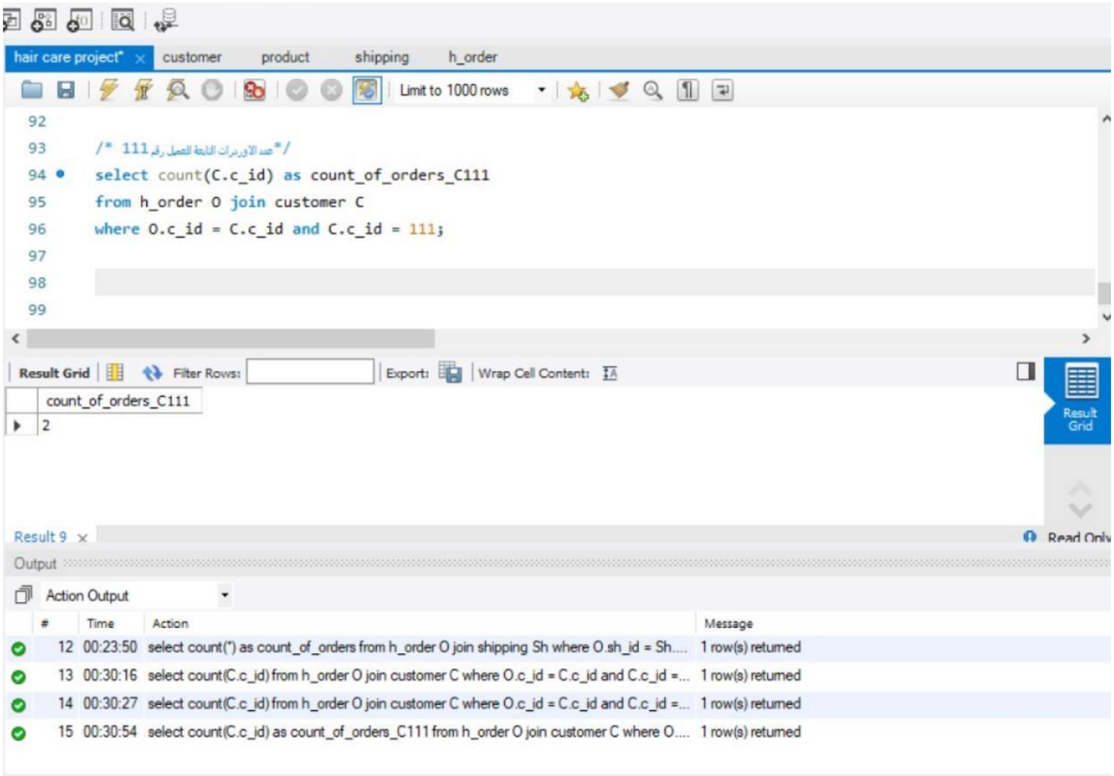
Our system allows businesses to evaluate their shipping efficiency .

For example :

understanding the count of orders linked to specific shipping methods helps in optimizing logistics and improving customer satisfaction.

8.

select count(C.c_id) as count_of_orders_C111
from h_order O join customer C
where O.c_id = C.c_id and C.c_id = 111;



Results Overview :

The system helps in monitoring customer engagement .

For example :

it counts the number of orders placed by the customer with ID 111.

System Benefits :

Our system provides insights into customer loyalty and purchasing behavior .

For example :

analyzing the count of orders for specific customers can guide targeted marketing efforts and improve customer retention strategies.

Conclusion

Summary of Achievements:

The Hair Care Product Online Store System designed a database that would include all sorts of components, like products, customers, orders, shipping, Admin . among others. It gives a easy in the process of purchasing products and finding desired products without having any issue in it. This system will let customers easily get through with every step for processing an order faster and getting their ordered item delivered successfully in good quality.

Lessons Learnt:

Through the project, we learnt the importance of very good planning and communication among the team members. Designing an accurate database schema reflecting the requirements of the user can save much hassle during implementation. We also came to understand that giving paramount importance to customer experience plays an important role in customer satisfaction and retention.

Recommendations for Future Projects:

In the future, extensive user testing should be conducted earlier in the process to obtain user feedback. This would help the system to meet users' expectations and be more easily navigated. The scalability of the system since its inception will prepare the system for scalability with the increase in the number of users. Finally, strong training resources for the end-users can further enhance their experience and reduce potential challenges when trying to navigate the system.